Fine Structure of Mixed-Base Tetration Phases: Certified Evaluation, a Universal Mode-Decay Law, and a Boundary Essential Singularity

Paper IV of the mixed-base tetration series

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July 2026

Abstract

For regular tetration branches T_b, T_d with $b, d > e^{1/e}$, the mixed-base phase $\Phi_{b,d}(\theta) = \lim_{n\to\infty}\big(\mathrm{slog}_b(T_d(n+\theta)) - (n+\theta)\big)$ is a small 1-periodic circle map whose existence, groupoid structure and C^1-diffeomorphism property were established in the companion papers [1, 2]. This paper resolves, partly by proof and partly by a high-precision measurement campaign, the fine structure of these phases, and turns them into a certified computational primitive for base change in tetration.

On the theoretical side we prove: an exact two-level entry and an a-posteriori error certificate for the logarithmic peel ladder, making every phase evaluation carry a proven error bound; a regularity dictionary "branch $C^m \Rightarrow$ Fourier modes $\asymp k^{-(m+2)}$" with an explicit seam-jump formula, calibrated on explicit C^1 branches (measured exponents 3.06 and 3.006); a first-order perturbation theory in the base mismatch $\varepsilon = \ln \log_b d$, in which the derivative of the phase at the diagonal $b = d$ is given by a response kernel solving a linear cohomological equation, explaining the empirical amplitude law $A_k(b, d) \approx C_k\, |\ln \log_b d|$; an exact hub decomposition $\mu_{b,d} = m(d) - m(b) + \Delta$ of the mean shifts with second- and third-order defect formulas validated to 10^{-15}; and a parabolic passage-time derivation of the edge law $m(b) \sim -\pi\sqrt{2\eta/e}\,(b-\eta)^{-1/2}$ as $b \downarrow \eta = e^{1/e}$, matching the measured coefficient to 2.3%.

On the empirical side, Fourier modes of the regular phases were measured to $k = 300$ after identifying and quantifying an instrument systematic of the underlying Kneser engine (usable band $\approx 2.3\, n_{\lim}$ theta harmonics). The decay profile $\sigma_{\mathrm{eff}}(k)$ is universal across all fourteen measured base pairs to three to four decimals; a direct exploration of the complexified phase locates an isolated *essential singularity* per period on the strip boundary, at pair-universal height $y_* \approx 0.055$ and with pair-universal strength, only its horizontal position $x_*(b, d)$ depending on the pair. Steepest-descent asymptotics link the singularity data to the measured four-parameter decay law and yield mode-count formulas for base-change tables (≈ 300 modes for 150 digits, ≈ 1400 for 500 digits). We close with a hybrid data format—mean-shift hub, Fourier head plus universal curve, certified peel ladder beyond—in which cross-base identities serve as proof residuals rather than heuristic checks.

1 Introduction

Regular tetration to a base $a > e^{1/e}$ is the increasing real-analytic solution T_a of

$$T_a(0) = 1, \qquad T_a(x+1) = a^{T_a(x)}, \tag{1}$$

*See the Authorship and AI Contribution Statement at the end of this manuscript: the research reported here was carried out in a directed human–AI mode, with the author acting primarily as research director and the mathematical development and numerical campaign largely produced by AI systems under his direction.

in the Kneser–Paulsen–Cowgill normalization [5, 6, 7, 8], with Abel inverse $A_a = T_a^{-1} = \mathrm{slog}_a$. Given two admissible bases, the mixed-base height map $F_{b,d} = A_b \circ T_d$ converts d-height into b-height. The first paper of this series [1] proved, conditionally on explicit channel and peeling-growth estimates for the regular branch, the *phase law*

$$\mathrm{slog}_b\big(T_d(x)\big) - x \;=\; \Phi_{b,d}(\{x\}) + o(1), \qquad x \to +\infty, \tag{2}$$

with $\Phi_{b,d}$ continuous and 1-periodic, decomposed as $\Phi_{b,d} = \mu_{b,d} + P_{b,d}$ into a mean shift and a centered periodic profile, and established the exact groupoid identity $F_{b,d} = F_{b,c} \circ F_{c,d}$ together with its nonlinear phase cocycle. The companion note [2] then de-conditionalized the smoothness assumptions: the lifted phase maps $\widetilde{H}_{b,d} = \mathrm{id} + \Phi_{b,d}$ form a groupoid of orientation-preserving circle homeomorphisms with $\widetilde{H}_{b,d}^{-1} = \widetilde{H}_{d,b}$, the inverse mean antisymmetry $\mu_{b,d} = -\mu_{d,b}$ holds exactly with no smoothness at all, and under a C^1 reading of the channel hypotheses the phase maps are C^1 circle diffeomorphisms with the exact derivative product identity

$$\widetilde{H}'_{b,d}(\theta) \;=\; \frac{\alpha\, T'_d(\theta)}{T'_b\big(\widetilde{H}_{b,d}(\theta)\big)} \prod_{j=1}^{\infty} \frac{\alpha\, T_d(j+\theta)}{T_b\big(j + \widetilde{H}_{b,d}(\theta)\big)} \;>\; 0, \qquad \alpha = \log_b d. \tag{3}$$

Three questions were left open, and they are precisely the questions that decide whether the phases can be used as a stable computational primitive for base change:

(Q1) *Regularity and mode decay.* Is $\Phi_{b,d}$ real-analytic on a strip ([1, Conj. 8.1]), merely C^∞, or something in between? Equivalently: how do the Fourier amplitudes $A_k = 2|\widehat{P}_{b,d}(k)|$ decay? The answer fixes the size of any tabulated representation of Φ.

(Q2) *Laws for the shift and the amplitudes.* The data of [1] hinted at $A_1 \approx C_1 |\ln \alpha|$ with a near-universal constant C_1, and at near-additive mean shifts. Are there provable first-order laws behind these observations, and what is the exact bookkeeping for $\mu_{b,d}$ across many bases?

(Q3) *Certified evaluation.* Can every evaluation of Φ, of its inverse, and of every reconstructed pair quantity carry a proven error bound suitable for use inside an automated verification loop?

This paper answers (Q3) affirmatively by proof, answers (Q2) by a perturbation theory whose predictions are confirmed to many digits, and answers (Q1) by a measurement campaign that replaces the dichotomy "strip versus C^∞" with a sharper picture: the phase extends holomorphically to a strip *punctured* at one point per period, where it has an isolated essential singularity of pair-universal type, whose height $y_* \approx 0.055$ (combined estimate $\sigma_* = 0.058 \pm 0.006$) is the strip half-width. All numerical claims come from a July 2026 measurement campaign against the author's high-precision Kneser engine (a fork of Levenstein's `fatou.gp` [9]); Appendix B documents the pipeline and the proof-residual gates under which the data were accepted.

1.1 Contributions

Certified algorithms (Section 3). We formalize the logarithmic peel ladder as an algorithm with an exact two-level entry (Lemma 3.1) and an a-posteriori certificate (Proposition 3.3): entering the ladder at a tower value $\geq M$ incurs a total phase error $\leq K d^{-M}$. Newton inversion of the phase map is uniformly well conditioned since $\widetilde{H}' \in [0.9971,\, 1.0030]$ across all measured pairs. We also identify, quantify and turn into a guard rule the dominant instrument systematic:

truncation of the engine's Kneser theta series at n_{lim} harmonics produces a deterministic mode floor at level $\approx e^{-2.95\, n_{\mathrm{lim}}}$ with onset $k \approx 2.3\, n_{\mathrm{lim}}$, independent of working precision.

Regularity dictionary (Section 4). For branch pairs of finite smoothness we prove the transport statement "branch C^m with seam jumps $\Rightarrow |\widehat{P}(k)| \asymp k^{-(m+2)}$" with an explicit two-seam interference structure (Propositions 4.1, 4.2). On the explicit C^1 branches of [2, Rem. 7.5(ii)] the measured exponents are 3.06 and 3.006, calibrating the pipeline's ability to detect polynomial signatures down to amplitudes 10^{-150}.

Mode decay and the boundary singularity (Sections 5, 6). Clean Fourier modes of the regular phases to $k = 300$ show a decay profile $\sigma_{\mathrm{eff}}(k)$ that is *universal* across all fourteen measured pairs. A direct complexified evaluation of the phase (a y-scan of the peel ladder) locates an isolated singularity per period at $z_* \approx x_* + 0.055\, i$; its strength and type are pair-universal to 0.01–0.3%, only the abscissa $x_*(b, d)$ varies. Steepest-descent asymptotics (Proposition 6.3) identify the measured four-parameter decay law $\ln A_k = \mathrm{const} + \gamma \ln k - 2\pi\sigma_* k - \sqrt{\pi c k}$ with an essential singularity of exponential type on the strip boundary, and two elementary consequences—a crossover scale (Lemma 6.4) and a digit-yield formula (Corollary 6.5)—explain the observed fit-window drift and give the mode counts needed for base-change tables.

First-order theory of amplitudes (Section 7). Differentiating the family of Abel functions in the base parameter $\beta = \ln\ln b$ yields a *response kernel* V_d solving the linear cohomological equation $V_d(d^y) - V_d(y) = -A_d'(d^y)\, y\, (\ln d)\, d^y$, which along the tower orbit collapses to the strikingly simple increment $-T_d/T_d'$. Under a base-differentiability hypothesis we prove $\Phi_{b,d} = \varepsilon\, G_d + O(\varepsilon^2)$ with $\varepsilon = \ln\alpha$ and an explicitly 1-periodic kernel profile G_d (Theorem 7.3), which upgrades the empirical amplitude law $A_k \approx C_k |\ln\alpha|$ to a derived first-order statement and identifies the hub slope $\frac{d}{d(\ln\ln b)}\mu_{e,b}\big|_{b=e} = \int_0^1 G_e$, measured as 2.191463277066. Ultra-near-diagonal experiments determine $C_1(e)$ to seventeen digits.

The mean-shift hub (Section 8). The exact antisymmetry of [2] makes the decomposition $\mu_{b,d} = m(d) - m(b) + \Delta_{b,c_*;d}$ exact with a single defect integral; we derive the third-order defect formula and validate the chain $\Delta^{(2)} + \Delta^{(3)}$ to $\sim 10^{-15}$ absolute. The hub function $m(b) = \mu_{e,b}$ was measured over 24 bases in $[1.46, 100]$; at the lower edge we *derive* the divergence law $m(b) \sim -\pi\sqrt{2\eta/e}\,(b-\eta)^{-1/2}$ from the parabolic bottleneck of the threshold base (Lemma 8.2), matching the fitted coefficient to 2.3%. PSLQ searches for closed forms of $m(2), m(3), m(5), m(10)$ and $C_1(e)$ against an extended constant basis were uniformly negative.

A hybrid data format (Section 9). We specify the format in which these results become an implementable base-change primitive: hub column, per-pair Fourier head plus universal decay curve for the working range $\lesssim 150$ digits, certified peel ladder beyond, and cross-base identities (mean antisymmetry to 10^{-38}, groupoid residuals) as proof residuals with thresholds derived from the certificate, not from experience.

1.2 Standing conditionality and provenance

All statements about the regular Kneser–Paulsen–Cowgill branch are conditional in exactly the sense of [2, Rem. 7.4]: they consume the real-channel and peeling-growth hypotheses of [1, Hyp. 2.2, 2.4], which remain to be derived from the branch construction itself. The toy-branch results of Section 4 are unconditional. Numerical values are quoted with the precision at which they passed the gates of Appendix B; the raw data, scripts and run metadata reside in the project repository.

2 Preliminaries

We keep the notation of [1, 2]. Throughout, $b, d > 1$ are admissible bases, T_a denotes the regular increasing branch normalized by (1) (frame (B1)–(B2) of [2]), $A_a = \mathrm{slog}_a$ its Abel inverse, $L_b = \log_b$, and

$$\alpha = \log_b d, \qquad D_n(\theta) = D_n^{b,d}(\theta) = L_b^{\circ n}\big(T_d(n+\theta)\big), \qquad \theta \in [0,1].$$

Under [1, Hyp. 2.2, 2.4], $D_n \to D$ uniformly, the phase law (2) holds with $\Phi_{b,d} = A_b \circ D - \mathrm{id}$, and $s_j := \inf_\theta T_d(j+\theta)$ satisfies the exact tower recursion $s_{j+1} = d^{s_j}$ ([2, Lem. 2.5]), so $\sum_j 1/s_j < \infty$ automatically. The σ-coordinates of [2, Lem. 6.1, 6.3] write the finite tails as

$$D_n'(\theta) = \alpha\, T_d'(\theta) \prod_{j=1}^{n-1}\big(1+\sigma_j^{(n)}(\theta)\big), \qquad \sigma_j^{(n)} = \frac{L_b(\alpha) + L_b\big(1+\sigma_{j+1}^{(n)}\big)}{\alpha\, T_d(j+\theta)}, \qquad \sigma_{n-1}^{(n)} = 0, \tag{4}$$

with the uniform tail bound $|\sigma_j^{(n)}| \le C_0/s_j \le \frac{1}{2}$ for $j \ge J_0$, where $C_0 = \big(|L_b(\alpha)| + \ln 2/\ln b\big)/\alpha$ ([2, Eq. (17)]). We write $\widehat{P}(k)$ for the complex Fourier coefficients of the centered profile $P = P_{b,d}$, $A_k = 2|\widehat{P}(k)|$ for the real mode amplitudes, and define the *local decay rate*

$$\sigma_{\mathrm{eff}}(k) \;=\; \frac{\ln A_k - \ln A_{k+1}}{2\pi}. \tag{5}$$

A strip-analytic P with half-width σ and no boundary structure would give $\sigma_{\mathrm{eff}}(k) \to \sigma$; a C^m profile with seam jumps gives $\sigma_{\mathrm{eff}}(k) \approx \frac{m+2}{2\pi k}$.

For an actual finite height $x = n + \theta$ we write $R_n(\theta) := F_{b,d}(n+\theta) - (n+\theta) - \Phi_{b,d}(\theta)$ for the *finite-height remainder*; $\|R_n\|_\infty \to 0$ at inverse-tower speed but $R_n \not\equiv 0$, so any claim that a phase table evaluates $F_{b,d}$ at finite x must carry this term in its error budget. The stopped-coordinate certificate of Paper III [3] bounds it by $2C_0'/T_b(n-2)$, which jumps from $\sim 5 \times 10^{-7}$ at $n = 5$ to $\sim 10^{-1{,}650{,}000}$ at $n = 6$ for $(e, 2)$: the switch height below which the direct engine must be used is tiny and pair-computable.

3 Certified evaluation: the peel ladder

The evaluation primitive behind every measurement in this paper is the logarithmic peel ladder. Its correctness rests on two exact identities and one hyper-fast tail bound; we record them in a form that yields a certificate per evaluation.

Lemma 3.1 (Exact two-level entry)**.** *For every $m \ge 0$ and $\theta \in [0,1]$ for which the first two peels stay in the real channel,*

$$L_b^{\circ 2}\big(T_d(m+2+\theta)\big) \;=\; \log_b \alpha + \alpha\, T_d(m+\theta). \tag{6}$$

Proof. $L_b\big(T_d(m+2+\theta)\big) = L_b\big(d^{T_d(m+1+\theta)}\big) = \alpha\, T_d(m+1+\theta)$, and then $L_b\big(\alpha T_d(m+1+\theta)\big) = \log_b \alpha + L_b\big(d^{T_d(m+\theta)}\big) = \log_b \alpha + \alpha\, T_d(m+\theta)$. □

Thus the ladder never needs a tower value beyond $T_d(m+\theta)$ in memory: the two topmost levels are absorbed exactly, and everything above them is a hyper-exponentially small σ-correction.

Algorithm 3.2 (Certified peel ladder)**.** Given (b,d), $\theta \in [0,1)$, a working threshold M and a landing window $\mathcal{W} = [w_-, w_+] \subset (0,\infty)$:

1. lift the tower from Taylor data on $[0,1]$ until $T_d(m+\theta) \ge M$;
2. enter with $W_0 := \log_b \alpha + \alpha\, T_d(m+\theta)$ (two levels absorbed exactly by Lemma 3.1);
3. peel $W \leftarrow L_b(W)$ until $W \in \mathcal{W}$, counting p peels;

4. return $\Phi(\theta) = A_b(W) + p - m - \theta$: since $\operatorname{slog}_b\big(T_d(m+2+\theta)\big) = A_b(W) + (2+p)$ after the two absorbed levels and p peels, subtracting the height $m+2+\theta$ collapses the bookkeeping, and a single slog_b call on the landing window closes the evaluation.

Proposition 3.3 (A-posteriori certificate)**.** *Assume the phase-law package and the quantitative channel of [2] for (b,d) (by Paper III [3], the growth hypothesis and finite-chain positivity hold unconditionally in stopped coordinates, and the limit channel reduces to one offset inequality per pair), and let J_0, C_0 be as in (4). If the ladder enters at a level m with $s := T_d(m+\theta) \geq M \geq s_{J_0}$, then its output $\Phi^{(m)}(\theta)$ satisfies*

$$\big|\Phi^{(m)}(\theta) - \Phi_{b,d}(\theta)\big| \;\leq\; K\, d^{-M},$$

where K depends only on C_0, on the Lipschitz constant of A_b on the landing window, and on the (finitely many) channel constants of the sub-window levels.

Proof sketch. Entering at level m is the same as evaluating the exact infinite object with all corrections σ_j at levels j above the entry set to zero. By [2, Lem. 6.3] the neglected corrections satisfy $|\sigma_j| \leq C_0/s_j$, and by the tower recursion $s_{j+1} = d^{s_j}$ the first neglected one already obeys $C_0/s_{m+1} = C_0\, d^{-s} \leq C_0\, d^{-M}$, with the remaining ones smaller than any fixed multiple of it. Transporting this perturbation down the ladder multiplies it by the product of the peel derivatives $1/(W_k \ln b)$, which is bounded by a constant on the compact channel below the entry, and the final slog_b is Lipschitz on the landing window. Summing the geometric-in-tower tail gives the stated bound. □

In the campaign the certificate was accompanied by an operational gate: adding one further explicit tower level changed no returned phase value by more than 10^{-120} at $M = 10^4$, consistent with the bound $C_0\, d^{-10^4}$.

Proposition 3.4 (Conditioned inversion)**.** *Let $\delta := \max_\theta |\Phi'(\theta)| \leq 2\pi \sum_k k\, A_k$. Then $\widetilde{H} = \mathrm{id} + \Phi$ is a C^1 diffeomorphism with $\widetilde{H}' \in [1-\delta, 1+\delta]$; the fixed-point iteration $\theta_{n+1} = u - \Phi(\theta_n)$ is a* global *contraction with factor δ (Newton acceleration is optional after a standard basin check); and the inverse phase requires no separate computation: $\widetilde{H}^{-1} = \widetilde{H}_{d,b}$ exactly [2, Thm. A]. The bounds $1-\delta$ are themselves certifiable from the mode table via $\delta \leq 2\pi \sum_k kA_k$ plus the tail envelope.*

Across all fourteen measured pairs, $\delta \leq 3.0 \times 10^{-3}$, i.e. $\widetilde{H}' \in [0.9971, 1.0030]$ (best pair $(e,3)$ with $\pm 2.3 \times 10^{-4}$, worst $(2,10)$ and $(10,2)$ with $\pm 3.0 \times 10^{-3}$): the phase maps are essentially perfectly conditioned.

3.1 An instrument law: the theta-truncation floor

The engine evaluates T_a and A_a through Kneser's construction with the 1-periodic theta mapping truncated at $n_{\lim}$ harmonics. This truncation is invisible at the level of individual function values at ordinary precision, but it is *not* invisible to high Fourier modes of Φ: it injects a deterministic, precision-independent periodic perturbation of the branch, which propagates linearly into Φ through the derivative transfer of (3). The campaign isolated this systematic cleanly:

Observation 3.5 (Mode floor). With $n_{\lim} = 30$ the mode amplitudes of $(e,2)$ freeze at $A \approx 2.6 \times 10^{-44}$ with onset $k \approx 56$; with $n_{\lim} = 60$ the floor moves to $A \approx 9.9 \times 10^{-83}$ with onset $k \approx 137$. The floor level is bit-identical between working precisions dps $= 140$ and 200 (agreement 10^{-25}), hence a systematic of the branch series, not rounding and not the ladder. The two measurements give the instrument law

$$\operatorname{floor}(n_{\lim}) \;\approx\; \mathrm{const} \cdot e^{-2.95\, n_{\lim}} = \mathrm{const} \cdot e^{-2\pi \cdot 0.47\, n_{\lim}}, \qquad k_{\text{floor}} \;\approx\; 2.3\, n_{\lim},$$

the rate $e^{-2\pi\cdot 0.47}$ per harmonic being the decay rate of the engine's Kneser theta series at its effective evaluation height.

The practical guard rule is immediate and is enforced in the pipeline: *modes beyond* $2.3\, n_{\lim}$ *are artifacts by definition*; for a target band $k \le k_{\max}$ one must initialize with $n_{\lim} \ge k_{\max}/2.3$ and choose the working precision from the floor law (both the rate and the measured prefactor, which sits about five orders below the pure rate extrapolation). Initialization costs scale accordingly ($n_{\lim} = 60$ at dps $= 200$: 56 s and 95 s for bases 2 and e; $n_{\lim} = 120$: 239 s and 196 s; one-time, disk-cached). A third calibration point at $n_{\lim} = 240$ (floor onset $k \approx 530$ against the predicted $2.3 \cdot 240 = 552$) confirms the law across an eightfold range. We emphasize the methodological point: this is the same failure mode—silent theta truncation—that historically corrupted reference values in the underlying engine; here it was detected *before* contaminating any conclusion, quantified, and converted into an instrument constant with a formula.

4 The regularity dictionary and its calibration

Question (Q1) is only meaningful if the measurement pipeline can distinguish regularity classes. The companion note provides the exact tool: by [2, Rem. 7.2], $D = T_b \circ \widetilde{H}$, so given branch regularity, regularity of Φ and of the limiting tail are equivalent, and *branch* regularity is transported into *phase* regularity with known bookkeeping. For finite smoothness this yields a quantitative dictionary.

Proposition 4.1 (Fourier asymptotics with seam jumps)**.** *Let $f : S^1 \to \mathbb{R}$ be C^m and piecewise C^{m+2}, with $f^{(m+1)}$ having jump discontinuities $J_1, \dots, J_r$ at $\tau_1, \dots, \tau_r \in S^1$. Then, as $k \to \infty$,*

$$\widehat{f}(k) \;=\; \frac{1}{(2\pi i k)^{m+2}} \sum_{i=1}^{r} J_i\, e^{-2\pi i k \tau_i} \;+\; o\Big(k^{-(m+2)}\Big).$$

Proof. Integrate by parts $m+1$ times; the boundary terms vanish by continuity of $f, \dots, f^{(m)}$ on the circle. One further integration by parts on each smooth piece produces the jump sum plus $(2\pi i k)^{-(m+2)}\widehat{f^{(m+2)}}(k)$ interpreted piecewise; the latter is $o(1)$ by Riemann–Lebesgue. □

Proposition 4.2 (Seam locations of the phase)**.** *Let the branches T_b, T_d be piecewise real-analytic with C^m seams at integer arguments (as for the explicit branches of [2, Rem. 7.5(ii)]). Then $\Phi_{b,d} \in C^m(S^1)$ is piecewise real-analytic, and within one period its seam set is*

$$\Sigma \;=\; \{0\} \;\cup\; \big\{\theta \in [0,1) :\; \widetilde{H}_{b,d}(\theta) \in \mathbb{Z}\big\},$$

i.e. the inner-branch seam at $\theta = 0$ together with the finitely many points where the phase map crosses an integer height of the outer branch. Consequently $\widehat{P}(k) = O(k^{-(m+2)})$, with the leading envelope governed by the finite seam sum of Proposition 4.1 (two-sided bounds $\asymp$ require a non-cancellation condition on that trigonometric sum) and with an interference oscillation between the seam contributions of Proposition 4.1, of beat period $\Delta k \approx 1/\mathrm{dist}(\Sigma)$ in k.

Proof sketch. For each θ, Algorithm 3.2 expresses $\Phi(\theta)$ through finitely many maps: the Taylor lift of T_d on $[0,1]$ (seams only where $m+\theta$ crosses $\mathbb{Z}$, i.e. $\theta = 0$), real logarithms (analytic on $(0,\infty)$), and one evaluation of A_b (seams where its argument crosses the seam values $T_b(\mathbb{Z})$, i.e. where $A_b(D(\theta)) = \widetilde{H}(\theta) \in \mathbb{Z}$). Away from Σ all maps are analytic; the C^m matching at the seams is the branch matching, transported by the chain rule; the seam count is finite because $\widetilde{H}$ is a homeomorphism of degree one. The Fourier statement is Proposition 4.1. □

Calibration measurement. On the explicit C^1 branches ($m = 1$; quadratic critical piece, seams with T'' jumps at integers) the dictionary predicts $|\widehat{P}(k)| \asymp k^{-3}$. Log–log fits over $k \in [4, 60]$ give decay exponents $p = 3.06$ for both orientations of $(e, 2)$ and $p = 3.006$ for $(2, 10)$, with the predicted interference oscillation clearly visible; for the toy pair $(e, 2)$, $\mu_{\text{toy}} \approx -1.1290$ places the outer seam at $\theta \approx 0.1290$, so the predicted beat period is $\Delta k \approx 1/0.129 \approx 7.8$, consistent with the observed oscillation. Two further pipeline gates were passed on the toy branches: $\mu_{\text{toy}}(e, 2) = -1.1290411574965$ agrees with the independent computation of [2, Rem. 7.5] up to -6.6×10^{-11}, which is exactly the expected N^{-3} aliasing error of the 512-point grid; and the inverse identity $\Phi_{2,e}(\widetilde{H}_{e,2}(\theta)) + \Phi_{e,2}(\theta) = 0$ holds to 5.7×10^{-101}. The pipeline therefore provably detects polynomial signatures; every regularity claim below is calibrated against this control.

5 Mode decay of the regular phases

For the regular branch the picture is entirely different from the toy control: no polynomial signature appears anywhere down to amplitudes 10^{-150}, and no geometric regime appears either.

Observation 5.1 (Concave decay, universal profile). For $(e, 2)$ at $n_{\text{lim}} = 120$, dps $= 190$, $N = 1024$ grid points, the modes are clean to $k \approx 290$ (no floor visible), with local rates

k	1	2	3	5	8	20	40	60	100	200	290
$\sigma_{\text{eff}}(k)$	0.740	0.567	0.487	0.404	0.344	0.260	0.2154	0.1946	0.1726	0.1484	0.1375

monotone and strictly concave in $\ln k$ throughout. The profiles $\sigma_{\text{eff}}(k)$ for $k \gtrsim 5$ agree to 3–4 decimals across *all* measured pairs—$(e, 2)$, $(2, e)$, $(2, 10)$, $(10, 2)$, $(e, 10)$, $(10, e)$, $(e, 3)$, $(3, e)$, $(e, 5)$, $(5, e)$, $(3, 5)$, $(5, 3)$, $(1.5, e)$, $(e, 1.5)$: the decay *shape* is a universal function $\Xi(k)$, and the pair enters essentially only through the amplitude $C_1|\ln \alpha|$ (plus a mild shape drift $e^{\delta k}$ with $|\delta| \sim 10^{-3}$).

Purely real-axis data to $k = 290$ admit two competing tail models: a strip of positive half-width $\sigma_\infty \in [0.05, 0.09]$ approached as $\sigma_{\text{eff}} = \sigma_\infty + \beta/\sqrt{k}$, or a stretched-exponential $\ln A_k \sim -ck^{0.79}$ with $\sigma_\infty = 0$, i.e. a Gevrey class far beyond all finite-smoothness classes. The first wins on $k \in [20, 140]$, the second on $k \in [140, 290]$; no two-parameter model fits globally. The dichotomy is resolved not by more modes but by going complex, which is the subject of the next section. What is already decided at this stage: [1, Conj. 8.1] *cannot* hold in its naive form with the first-mode rate ($\sigma \approx 0.74$)—the strip, if any, is an order of magnitude narrower—and the phase is not of any finite smoothness class.

For applications the dispute is immaterial in the working range: the measured digit yield $D(k) = \log_{10}(1/A_k)$ is

$$D(60) = 47, \qquad D(100) = 67, \qquad D(120) = 76, \qquad D(300) \approx 148,$$

identical for both models where measured, and both extrapolate 500 digits to $k \approx 1300$–1700 modes (Corollary 6.5).

6 The boundary singularity

6.1 Direct complexified evaluation

The peel ladder complexifies: for $\theta + iy$ with $y > 0$ the tower, the peels and the final slog_b all extend, as long as every intermediate value avoids the logarithmic cut $(-\infty, 0]$ and the solver tracks a consistent branch. The campaign implemented this as a bottom-up solver for

$\mathrm{sexp}_b(J+z+\varphi) = \alpha\, T_d(J+z)\cdot(1+\sigma\text{-closure})$ with the entry level J chosen per point, secant iteration with damping, continuation along paths in y and x (never cold evaluation near the critical zone), and, crucially, *mid-strip validation*: away from the critical zone the solver agrees with the $k \le 300$ Fourier reconstruction to below the per-point closure bounds (typically 10^{-20}–10^{-60}), so both instruments certify each other where both are valid.

Observation 6.1 (Breach geometry). For $(e,2)$ the solver is regular on the whole strip up to $y \approx 0.16$ *except* in a single slanted sector per period: at $y = 0.060$ the failure zone is $x \in [1.004, 1.022]$, widening through $[0.982, 1.012]$ at $y = 0.065$, $[0.964, 1.000]$ at 0.070, $[0.91, 0.98]$ at 0.085, $[0.879, \ge 0.95]$ at 0.095, with center slope $dx/dy \approx -2.8$ and apex

$$z_* \;\approx\; (1.02 \pm 0.02) \;+\; (0.055 \pm 0.005)\, i \pmod 1.$$

Inside the zone the level-J system exhibits quantized sheet slips and non-convergence (root density), with closure bounds down to 10^{-305}—excluding any truncation artifact. Off the zone, lines continue smoothly far beyond the apex height (measured cleanly to $y \approx 0.164$), with indications of a second structure at $y \approx 0.17$–0.22, $x \approx 0.6$–0.7, left to future work. The mode phases corroborate the location: the drift of $\arg \widehat{P}(k)$ defines $x_*(k)$ moving $0.47 \to 0.65$ over $k \le 280$ with a $1/\sqrt{k}$ approach toward $x_* \approx 0.94$–1.0.

Observation 6.2 (Universality of the singularity). At fixed height $y = 0.085$ the failure zones of three structurally different pairs are congruent: $(e,2)$: $[0.91, 0.98]$; $(3,5)$: $[1.01, 1.08]$ (a rigid shift by $+0.100$, exactly the phase-fit shift); $(2,10)$: $[0.650, 0.710]$—same height, same width (≈ 0.06–0.07), same shape; only the abscissa $x_*(b,d)$ is pair-dependent (centers $\approx 0.945/1.045/0.680$; the near-round values invite scrutiny). Structure fits over 220 modes (amplitudes and phases, $k \in [60, 280]$) with the one-singularity saddle model of Proposition 6.3 give

$$\begin{aligned} (e,2):&\quad \sigma_* = 0.0789, \quad c = 59.78, \quad \gamma = 9.47 \quad \text{(rms 0.005 over 220 modes)};\\ (3,5):&\quad \sigma_* = 0.0791, \quad c = 59.77, \quad \gamma = 9.59, \end{aligned}$$

i.e. σ_* pair-equal to 2.4×10^{-4} and c to 0.013%; the phase fit yields in addition a complex strength with arg-parameter $\chi \approx 0.68$ rad, pair-universal to 0.3%.

6.2 Saddle-point asymptotics

The measured four-parameter law is exactly the Fourier signature of an isolated essential singularity of exponential type sitting on the strip boundary.

Proposition 6.3 (Singularity $\Rightarrow$ decay law)**.** *Let P be real on $\mathbb{R}$, 1-periodic, holomorphic on $\{|\mathrm{Im}\,\theta| < \sigma_*\}$ except for isolated singularities per period at $\theta_* = x_* + i\sigma_*$ and $\bar\theta_*$, where locally*

$$P(\theta) \;=\; h(\theta)\,\exp\!\Big(\frac{v}{\theta - \bar\theta_*}\Big), \qquad h(\theta) \sim h_0\,(\theta - \bar\theta_*)^{-\rho},$$

for some $v \in \mathbb{C}\setminus\{0\}$, $\rho \in \mathbb{R}$, and suppose the contour can be deformed to the boundary away from the singularity. Then, as $k \to +\infty$,

$$\ln\big|\widehat{P}(k)\big| \;=\; \text{const} + \gamma \ln k - 2\pi\sigma_* k - \sqrt{\pi c\, k} + o(\sqrt{k}), \qquad \arg \widehat{P}(k) \;=\; -2\pi k\, x_* + O(\sqrt{k}),$$

where the strength satisfies $8|v|\cos^2\Psi = c$ with Ψ the steepest-descent phase determined by $\arg v$, and $\gamma = \rho/2 - 3/4$ collects the algebraic prefactor ($h \sim h_0 s^{-\rho}$) and the $O(k^{-3/4})$ saddle-width contribution. In particular, for real negative $v = -a$ one has $\Psi = \pi/4$ and $a = c/4$.

Proof sketch. For $k > 0$ push the contour of $\widehat{P}(k) = \int_0^1 P(\theta)e^{-2\pi ik\theta}\,d\theta$ to $\operatorname{Im}\theta = -\sigma_*$; the smooth part contributes $O(e^{-2\pi\sigma_* k})$ with bounded prefactor, and the dominant term comes from the neighborhood of $\bar\theta_*$. Writing $\theta = \bar\theta_* + s$, the local integral is $e^{-2\pi ik\bar\theta_*}\int h\,\exp\big(v/s - 2\pi iks\big)\,ds$ with saddle $s_* = \sqrt{iv/(2\pi k)}$ and saddle value $\varphi(s_*) = -4\pi iks_*$, so $|e^{\varphi(s_*)}| = \exp\big(-\sqrt{8\pi|v|k}\,\cos\Psi\big)$ with $\cos\Psi$ fixed by the admissible steepest-descent branch of $\sqrt{iv}$; the saddle width $|s_*| \propto k^{-1/2}$ and the prefactor $h_0 s_*^{-\rho}$ produce the algebraic factor k^γ. Matching $\sqrt{8\pi|v|}\cos\Psi = \sqrt{\pi c}$ gives the strength relation; standard steepest-descent estimates (e.g. [10, Ch. 4]) control the error terms. The check for real $v = -a$: the saddle direction gives $\cos\Psi = 1/\sqrt{2}$ and $|e^{\varphi(s_*)}| = e^{-2\sqrt{\pi ak}}$, i.e. $a = c/4$. □

With the fitted $c \approx 59.8$ this puts the singularity strength at $|v|\cos^2\Psi = c/8 \approx 7.5$, with $\arg v$ determined by the measured $\chi \approx 0.68$ rad. Two elementary consequences organize the real-axis phenomenology completely.

Lemma 6.4 (Crossover scale)**.** *In the model of Proposition 6.3 the increments of $-\ln A_k$ are dominated by the $\sqrt{k}$ term for $k < k_c$ and by the linear term for $k > k_c$, where*

$$k_c \;=\; \frac{c}{16\pi\sigma_*^2}.$$

Proof. Equate the derivatives $2\pi\sigma_*$ and $\frac{1}{2}\sqrt{\pi c/k}$. □

For $c = 59.8$: $k_c \approx 190$ if $\sigma_* = 0.079$, but $k_c \approx 390$ if $\sigma_* = 0.055$. This single number explains the entire fit phenomenology: window fits below k_c systematically *overestimate* σ_* (the $\sqrt{k}$ term masquerades as a larger linear rate), which is exactly the observed downward drift of subwindow estimates ($0.0822 \to 0.0746$ as the window moves up), converging toward the directly measured apex height 0.055 ± 0.005. The discrimination experiment was subsequently run: an $n_{\lim} = 240$ campaign (dps $= 260$, $N = 2048$) yields clean modes to $k \approx 520$ (floor onset ≈ 530, a third confirmation of the instrument law of Observation 3.5), and the sequence of sliding three-parameter window fits converges with a clean $1/w$ law, $\sigma_*(w) = 0.062 + 2.83/w$ reproducing all windows to $\pm 5\times10^{-4}$, extrapolating to $\sigma_*^\infty = 0.062 \pm 0.003$. Note that no single mode value discriminates (holding the subleading coefficients fixed misreads $\sigma_{\text{eff}}(500) = 0.1237$ as $\sigma_* \approx 0.078$); only the window-sequence extrapolation separates the hypotheses. Combining with the apex,

$$\boxed{\sigma_* \;=\; 0.058 \pm 0.006}$$

and the fitted 0.079 is *excluded* as an asymptotic value: it is the pre-asymptotic effective rate of windows below the crossover.

Corollary 6.5 (Digit yield and mode counts)**.** *In the fitted model the digit yield $D(k) = \log_{10}(1/A_k)$ satisfies $D(k)\ln 10 = -\ln C - \gamma\ln k + 2\pi\sigma_* k + \sqrt{\pi ck}$. Calibrating the single constant $\ln C$ at the measured $D(300) = 148$ reproduces the measured yields to ±1 digit ($D(60) = 46.0$ vs. 47; $D(100) = 65.9$ vs. 67; $D(120) = 75.1$ vs. 76) and gives $D = 500$ at $k \approx 1400$ (central value; 1300–1700 across the admissible parameter range), i.e. $n_{\lim} \approx 600$–750 with a one-time initialization of 1–2 hours per base.*

Conjecture 6.6 (Universal singularity data)**.** *For all admissible pairs (b, d), $b, d > e^{1/e}$, the phase $\Phi_{b,d}$ extends holomorphically to the punctured strip $\{|\operatorname{Im}\theta| < \sigma_*\} \setminus (\{\theta_*\} + \mathbb{Z})$ with an isolated essential singularity of exponential type at $\theta_* = x_*(b, d) + i\sigma_*$; the height σ_* is independent of the pair; the normalized strength $|v|$ and phase $\arg v$ are pair-independent to leading order (the diagonal kernels of Section 7 carry a smooth inner-base dressing of 0.4% at $k = 2$ growing to $\sim 9\%$ at $k = 8$, so exact universality can hold at most for the height and the singularity type), and $x_*(b, d)$ equals the accumulation point of the mode phases, $x_* = -\lim_k \arg\widehat{P}(k)/(2\pi k)$ along the drift-corrected sequence. In particular [1, Conj. 8.1] holds in the punctured form, with universal boundary data.*

The universality of the profile $\Xi(k)$ (Observation 5.1) and of the fitted (σ_*, c, χ) (Observation 6.2) is strong evidence; the linear-response theory of the next section reduces the conjecture, for near-diagonal pairs, to a statement about a single one-parameter family of kernels.

7 First-order theory: the response kernel

The empirical amplitude law $A_k(b,d) \approx C_k\,|\ln\alpha|$, with C_1 constant to $\pm 2.2\%$ over the full measured range $\ln\ln b \in [-0.90, 0.83]$, calls for perturbation theory in the mismatch parameter

$$\varepsilon \;:=\; \ln\alpha \;=\; \ln\ln d - \ln\ln b,$$

which vanishes exactly on the diagonal $b = d$, where $\Phi_{d,d} \equiv 0$. The natural base coordinate is $\beta := \ln\ln b$.

Hypothesis 7.1 (Base differentiability). *The family $b \mapsto (T_b, A_b)$ of regular branches is C^1 in β near $\beta_0 = \ln\ln d$, locally uniformly on compacts of the respective domains, with continuous derivative $V_d(y) := \partial_\beta A_b(y)\big|_{b=d}$; moreover $\partial_\beta \Phi_{b,d}(\theta)\big|_{b=d} = \lim_n V_d\big(T_d(n+\theta)\big)$ with the interchange of limit and derivative uniform in θ.*

For the intended branches this is supported by Paulsen's holomorphy of the construction in the base parameter [8] and is numerically evident; we state it as a hypothesis in the same conditional spirit as [2, Rem. 7.4].

Lemma 7.2 (Cohomological equation). *Under Hypothesis 7.1, differentiating the global Abel equation $A_b(b^y) = A_b(y) + 1$ in β at $b = d$ gives, for all $y \in \mathbb{R}$,*

$$V_d(d^y) - V_d(y) \;=\; -A_d'(d^y)\, y\, (\ln d)\, d^y, \tag{7}$$

and along the tower orbit $y = T_d(x)$ the right-hand side collapses:

$$V_d\big(T_d(x+1)\big) - V_d\big(T_d(x)\big) \;=\; -\frac{T_d(x)}{T_d'(x)}. \tag{8}$$

Proof. $\partial_\beta b^y = y(\ln b) b^y$ since $d\ln b/d\beta = \ln b$; evaluate at $b = d$ to get (7). For (8) use $A_d'(z) = 1/T_d'(A_d(z))$, $A_d(d^y) = x+1$, and $T_d'(x+1) = (\ln d) T_d(x+1) T_d'(x)$: $A_d'(d^y)\, y (\ln d) d^y = \dfrac{T_d(x)\,(\ln d)\,T_d(x+1)}{(\ln d)\,T_d(x+1)\,T_d'(x)} = \dfrac{T_d(x)}{T_d'(x)}$. ☐

Since $T_d'(n) \gtrsim (\ln d)^n \prod_{i<n} T_d(i)$ by iterating the derivative transfer, the increments (8) decay at inverse-tower speed, and the following limit is hyper-fast convergent.

Theorem 7.3 (Response kernel and first-order phase). *Assume the phase-law package for pairs near the diagonal and Hypothesis 7.1. Define*

$$S_d(\theta) \;:=\; \sum_{j\ge 0} \frac{T_d(j+\theta)}{T_d'(j+\theta)} \qquad (\textit{hyper-exponentially convergent}), \qquad G_d(\theta) \;:=\; S_d(\theta) - V_d\big(T_d(\theta)\big).$$

Then G_d is exactly 1-periodic and continuous, and as $b \to d$,

$$\Phi_{b,d}(\theta) \;=\; \varepsilon\, G_d(\theta) \;+\; O(\varepsilon^2) \qquad \textit{uniformly in } \theta.$$

Proof. Write $v(x) := V_d(T_d(x))$. By (8), $v(n+\theta) = v(\theta) - \sum_{j=0}^{n-1} T_d(j+\theta)/T_d'(j+\theta)$, so $\lim_n v(n+\theta) = v(\theta) - S_d(\theta) = -G_d(\theta)$ exists with hyper-fast convergence. By Hypothesis 7.1, $\partial_\beta \Phi_{b,d}(\theta)\big|_{b=d} = \lim_n v(n+\theta) = -G_d(\theta)$, and since $\varepsilon = \text{const} - \beta$ at fixed d, $\partial_\varepsilon \Phi\big|_{b=d} = G_d$; Taylor's theorem with the locally uniform derivative gives the expansion. Periodicity: $G_d(\theta+1) - G_d(\theta) = \big(S_d(\theta+1) - S_d(\theta)\big) - \big(v(\theta+1) - v(\theta)\big) = -\frac{T_d(\theta)}{T_d'(\theta)} - \big(-\frac{T_d(\theta)}{T_d'(\theta)}\big) = 0$, using (8) once more. ☐

Corollary 7.4 (First-order laws). *With $\widehat{G}_d(k)$ the Fourier coefficients of G_d:*

(i) Amplitude law. $A_k(b,d) = 2|\widehat{G}_d(k)|\,|\varepsilon| + O(\varepsilon^2)$*; hence* $C_k(d) = 2|\widehat{G}_d(k)|$*, and the observed near-universality of* C_1 *in* d *is the statement that* $d \mapsto 2|\widehat{G}_d(1)|$ *is nearly flat.*

(ii) Mean law and hub slope. $\mu_{b,d} = \varepsilon\,\overline{G}_d + O(\varepsilon^2)$ *with* $\overline{G}_d = \int_0^1 G_d$*; combined with the exact antisymmetry [2, Thm. B], the hub function* $m(b) = \mu_{e,b}$ *satisfies*

$$\frac{dm}{d(\ln\ln b)}\bigg|_{b=e} = \overline{G}_e.$$

(iii) Universal diagonal shape. *The centered profile shape at the diagonal is* $G_d - \overline{G}_d$*; its mode ratios are the* $\varepsilon \to 0$ *limits of the measured pair ratios.*

Measurements. Ultra-near-diagonal runs (d', e) with $\ln d' = 1 + \varepsilon$, $\varepsilon \in \{10^{-12}, 10^{-15}\}$, extract the first-mode ripple ($\sim 4 \times 10^{-19}$) cleanly above the engine floor and determine

$$\begin{aligned} C_1(e) &= 2|\widehat{G}_e(1)| = 0.00038859729244516473 \quad (\pm 1 \text{ ulp at 17 digits}), \\ \overline{G}_e &= 2.191463277066, \end{aligned}$$

the latter measured as the hub slope $dm/d(\ln\ln b)|_{b=e}$; the diagonal shape ratios are $A_2/A_1 = 0.0093355030$, $A_3/A_1 = 2.66505 \times 10^{-4}$, $A_4/A_1 = 1.26473 \times 10^{-5}$, and the diagonal first-mode phase is $\varphi_1 = 1.33536363153$ rad—close to, but clearly distinct from, $\operatorname{Im} L_e = 1.33724$ (L_e the principal fixed point of e^z). Near-diagonal runs at ± 0.02 give the kernel amplitude curve

$$C_1(d)\ [\times 10^{-4}]: \quad 3.953\ (d{=}2), \quad 3.886\ (e), \quad 3.868\ (3), \quad 3.793\ (5), \quad 3.720\ (10),$$

nearly affine in $\ln\ln d$ (slope $\approx -0.19 \times 10^{-4}$), with the quadratic ($\varepsilon^2$) deviations resolved at $\sim 9 \times 10^{-6}\,\varepsilon$. These values were subsequently *derived*: the selection principle for V_d is formulated, proved unique, and proved solvable in Paper V of this series [4], and its constructive Riemann–Hilbert realization reproduces the kernel family engine-free to 30–34 digits for $d \in \{2, e, 3, 5, 10\}$, in mutual agreement with ε-stencil differentiation of the branch family. Two independent routes thereby exposed an error in a third (complex-step references at the extreme bases $d = 2$ and $d = 10$, reliable only to ≈ 20 resp. ≈ 15 digits: the engine floor at $|\lambda(2)| = 1.23$ near the parabolic edge, and the narrowed theta strip at $\ln 10$); all kernel values quoted here are confirmed by the surviving routes. The second-order (in ε) kernels are measured as well, e.g. $\overline{H}_e = 1.66735420625407$ with the $\ln\ln$-exact node parametrization, and the two-term Taylor law $\widehat{P}_k = \varepsilon\widehat{G}_k + \varepsilon^2\widehat{H}_k$ reproduces the mode heads of *all* measured pairs to third-order-small residuals even at $\varepsilon = 1.20$.

8 The mean-shift hub

8.1 Exact decomposition and higher-order defects

Proposition 8.1 (Exact hub). *Fix a hub base c_* and set $m(b) := \mu_{c_*,b}$. Then for all admissible b, d,*

$$\mu_{b,d} = m(d) - m(b) + \Delta_{b,c_*;d}, \qquad \Delta_{b,c_*;d} = \int_0^1 P_{b,c_*}\big(\{\theta + \Phi_{c_*,d}(\theta)\}\big)\,d\theta,$$

exactly: the identity combines the integrated cocycle [1, Eq. (39)–(40)] with the exact antisymmetry $\mu_{b,c_*} = -\mu_{c_*,b}$ [2, Thm. B]. Expanding the defect in the small profiles,

$$\Delta^{(2)} = \sum_{k\neq 0} 2\pi ik\,\widehat{P}_{b,c_*}(k)\,e^{2\pi ik\mu_{c_*,d}}\,\widehat{P}_{c_*,d}(-k), \qquad \Delta^{(3)} = \frac{1}{2}\sum_{k\neq 0}(2\pi ik)^2\,\widehat{P}_{b,c_*}(k)\,e^{2\pi ik\mu_{c_*,d}}\,\widehat{P^2_{c_*,d}}(-k),$$

with $\left|\Delta - \Delta^{(2)} - \Delta^{(3)}\right| \le \frac{1}{6}\,\|P'''_{b,c_*}\|_\infty\,\|P_{c_*,d}\|_\infty^3$, *i.e. the error carries two more powers of the (tiny) amplitudes than* $\Delta^{(2)}$ *itself.*

Proof. Exactness is the cited combination. For the expansion, Taylor $P_{b,c_*}(\theta+\mu+P_{c_*,d}) = P_{b,c_*}(\theta+\mu) + P'_{b,c_*}(\theta+\mu)P_{c_*,d} + \frac{1}{2}P''_{b,c_*}(\theta+\mu)P^2_{c_*,d} + O(\|P\|^3\|P'''\|)$, integrate, note the first term has mean zero, and convert the two integrals by Parseval, $\int fg = \sum \widehat{f}(k)\widehat{g}(-k)$, with $\widehat{f'}(k) = 2\pi i k \widehat{f}(k)$ and the shift factor $e^{2\pi i k\mu}$. □

The hub economics are the point: N bases require one hub column of N values (plus low-k mode heads) instead of N^2 pair runs. With $c_* = e$ the campaign validated the chain at the level the formulas predict: the ratios $\Delta^{(2)}/\Delta_{\text{measured}}$ are 0.999816, 0.999900, 0.999651, 1.000074 for $(2,10)$, $(10,2)$, $(3,5)$, $(5,3)$ (residual = the $\Delta^{(3)}$ term), and including $\Delta^{(3)}$ the ratios become 1.0027, 1.0010, 1.0005, 0.9999 respectively—the truncated hub chain $\mu = m(d) - m(b) + \Delta^{(2)} + \Delta^{(3)}$ has the scale the remainder bound predicts: from the displayed ratios, the largest tested defect $(2,10)$ closes to about 1.7×10^{-11} absolute and the closer pair $(3,5)$ to about 1.5×10^{-13}; the full-precision chain in the repository carries the reconstruction further. As a scaling check, $\Delta(3,5) = -2.96\times10^{-10}$ is $20\times$ smaller than $\Delta(2,10) = 6.4\times10^{-9}$, as befits the closer base pair.

8.2 The hub function: values, edge law, large bases

High-precision hub values (regular branch; 150–180 digits available for $m(2)$, $\sim$ 170 for $m(3), m(5), m(10)$; leading digits shown):

$$\begin{aligned}
m(1.5) &= -9.776277436687167726082341422480654566832\ldots\\
m(2) &= -1.128403776628924009879217902036426267112\ldots\\
m(3) &= +0.192422293722432083208244406689281327790 3\ldots\\
m(5) &= +0.771257296154475876380441053974662638104 3\ldots\\
m(10) &= +1.136557055713326440899375967586836969056\ldots
\end{aligned}$$

together with a 24-point raster over $b \in [1.46, 100]$ at 40 digits. (The value of $\mu_{e,2}$ corrects the twelve-digit Table 1 entry of [1], whose known 9.3×10^{-9} error is exactly the inverse-sum residual reported there; the regenerated table appears in Appendix A. Pairwise mean antisymmetry now holds to $\le 10^{-39}$.)

At the lower edge the hub diverges, and the divergence law can be derived: it is the passage time through the parabolic bottleneck of the threshold base.

Lemma 8.2 (Parabolic edge law). *Let* $\eta = e^{1/e}$ *and* $b \downarrow \eta$. *To leading order in the continuum (Fatou coordinate) approximation of the bottleneck,*

$$m(b) \;=\; \mu_{e,b} \;=\; -\pi\sqrt{\frac{2\eta}{e}}\,(b-\eta)^{-1/2}\,\bigl(1+o(1)\bigr), \qquad \pi\sqrt{2\eta/e} \;=\; 3.23907\ldots,$$

with a subleading correction of order $\ln(b-\eta)$.

Derivation. At $b = \eta$ the map $t \mapsto b^t$ has the parabolic fixed point $t = e$ with multiplier 1. Write $b = \eta e^\delta$, $\delta = \ln(b/\eta) = (b-\eta)/\eta\,(1+O(b-\eta))$, and $t = e + w$. Then

$$b^t = \exp\bigl((e+w)(1/e+\delta)\bigr) = e\cdot\exp\bigl(w/e + e\delta + w\delta\bigr) = e + w + e^2\delta + \frac{w^2}{2e} + O(w\delta, w^3),$$

so the orbit obeys $w_{n+1} = w_n + a + Bw_n^2 + \cdots$ with $a = e^2\delta$, $B = 1/(2e)$. The continuum crossing time of such a bottleneck is $\int_{-\infty}^{\infty} dw/(a+Bw^2) = \pi/\sqrt{aB} = \pi\sqrt{2/(e\delta)} = \pi\sqrt{2\eta/e}\,(b-$

$\eta)^{-1/2}(1+o(1))$. A b-tower therefore lags a comparison e-tower by this many height units while traversing values near e, and the lag is exactly what $\mu_{e,b} = \lim(\mathrm{slog}_e T_b - \mathrm{id})$ measures (with the sign of a delay). The logarithmic subleading term is the standard second-order correction of parabolic passage (Fatou coordinates; see [11, §10]), and the derivation is made rigorous by the same machinery; we use only the leading order here. □

The measured fit on $b \in [1.46, 1.6]$ is $m(b) = -A(b-\eta)^{-1/2} + B\ln(b-\eta) + C$ with $A = 3.3148$, $B = -0.398$, $C = 3.163$ (rms 0.004): the ratio $A/\pi\sqrt{2\eta/e} = 1.023$ confirms the derived coefficient at the 2.3% level, the discrepancy being consistent with the fit window not yet asymptotic (the prediction is that $A \to 3.2391$ as the window closes on η; bases below 1.46 were not yet initialized). Notably the construction remains numerically robust arbitrarily close to the edge: base 1.5 (only 3.8% above η) initializes in 1.2 s and shows the same universal decay profile, with $C_1 = 3.97 \times 10^{-4}$ drifting mildly above the bulk band. At the other end, $m(b) - \ln\ln b$ decreases from 0.31 at $b = 8$ to 0.073 at $b = 100$; the natural conjecture $m(b) = \ln\ln b + o(1)$ (with rate $O(1/\ln b)$?) is left open.

8.3 Closed forms: negative evidence

PSLQ searches at 150 digits and coefficient bounds 10^6–10^8 found *no* integer relations for $m(2)$ against $\{1, \ln 2, 1/\ln 2, \ln\ln 2, \ln 2/(1-\ln 2), e, \gamma\}$, nor for $m(2), m(3), m(5), m(10)$ against an extended basis including η, π, e, logarithms, iterated logarithms, and the real and imaginary parts of the principal fixed points L_2, L_e; no relations hold among the m-values themselves, and short-precision probes of $C_1(e)$ and of the hub slope were likewise null. Together with the seventeen-digit $C_1(e)$, this is meaningful negative evidence that the phase constants live outside the standard constant ring—consistent with their definition through the Kneser uniformization, and consistent with the diagonal phase φ_1 landing *near* but not *at* $\operatorname{Im} L_e$.

9 A hybrid data format for base-change computation

The results above pin down the format in which the phases become a stable computational primitive inside a high-precision tetration engine. The design principle is that the two tail models of Section 5 coincide throughout the tabulated range, so no open question blocks implementation; and that every reconstructed quantity is guarded by an identity that is a *theorem*, so residuals are error meters rather than plausibility checks.

1. **Phase hub.** The groupoid gives the exact representation $\widetilde{H}_{b,d} = U_b^{-1} \circ U_d$ with $U_a := \widetilde{H}_{c_*,a}$: one phase map per supported base reconstructs every ordered pair, with stable error propagation since $\inf U_a' \geq 1 - \delta$ is certifiable from the mode table. Direct pair tables are optional caches, not fundamental data.

2. **Hub column.** Store $m(b) = \mu_{e,b}$ to full precision for each supported base (N values), plus the low-k mode heads. Reconstruct any pair mean as $\mu_{b,d} = m(d) - m(b) + \Delta^{(2)} + \Delta^{(3)}$ (Proposition 8.1); the chain closes to 10^{-11}–10^{-13} absolute at the displayed precision (Proposition 8.1), and a direct pair run remains available beyond that.

3. **Fourier table for the working range.** For target precisions $\lesssim 150$ digits ($k \leq 300$): per pair a low-k head ($k \leq 10$–20 complex coefficients) plus the universal curve, $A_k \approx C_1 |\ln\alpha|\, \Xi(k)\, e^{\delta k}$ (three parameters per pair, $|\delta| \sim 10^{-3}$).

4. **Certified ladder beyond.** From $\sim$ 150–200 digits upward, evaluate pointwise by Algorithm 3.2 with the certificate of Proposition 3.3: cost one sexp_d plus one slog_b per evaluation at arbitrary precision. Inversion by Proposition 3.4; the reverse map is never

computed separately. A 500-digit Fourier table is possible (Corollary 6.5: $k \approx 1400$, $n_{\lim} \approx 600$–750, 1–2 h one-time initialization per base) but pays off only under many evaluations per pair.

5. **Instrument guard.** Enforce $k_{\max} \leq 2.3\, n_{\lim}$ (Observation 3.5); choose working precision from the floor law including the measured prefactor.

6. **Gates as proof residuals.** Mean antisymmetry $\mu_{b,d} + \mu_{d,b} = 0$ (exact by [2, Thm. B]; achieved $\leq 10^{-38}$), groupoid residual $\Phi_{d,b} \circ \widetilde{H}_{b,d} + \Phi_{b,d} = 0$ (achieved 2.4×10^{-34}, limited only by table storage precision), and the truncation gate of Proposition 3.3 (one extra tower level, $< 10^{-120}$). Thresholds derive from the certificate constants, not from experience.

10 Discussion and open problems

The phase program set out in [1] asked whether mixed-base tetration heights are related by a small, structured, computable correction. At the level of practice the answer is now complete: the correction is computable at arbitrary precision with certificates, its inverse is free, its mean obeys an exact hub calculus, and its Fourier representation is governed by one universal curve plus a handful of per-pair constants. At the level of structure, the surprise of the campaign is how much is *universal*: the decay profile, the singularity height, strength and type are shared across every measured pair, including pairs with no base in common; all pair dependence concentrates in one amplitude $C_1|\ln \alpha|$, one abscissa $x_*(b,d)$, and one hub potential $m(b)$. We record the open problems in the order we consider them decisive.

1. **The residual σ_* decimal and the fit prefactor.** The window-sequence verdict $\sigma_* = 0.058 \pm 0.006$ (Section 6) should be sharpened by sector-resolved apex sweeps at $\Delta x = 5 \times 10^{-4}$, closing the seam subtlety by evaluating the zone in the adjacent cell via $D(1+\theta) = b^{D(\theta)}$; and the implausibly large effective exponent $\gamma \approx 9.5$ (i.e. $\rho \approx 20$ via $\gamma = \rho/2 - 3/4$) quantifies that the fitted prefactor is still pre-asymptotic—the asymptotic (σ_*, c, γ) triple remains to be pinned.

2. **Prove the punctured strip.** The complexified σ-recursion (4) bounds all but finitely many levels on any substrip, so the channel can only break at finitely many explicit levels: Conjecture 6.6 reduces to controlling those, in the spirit of [2, Rem. 7.3], plus a local analysis at the breach. The measured breach geometry (Observation 6.1) tells us exactly where.

3. **Extend the kernel theorem.** The first-order kernel is now characterized and constructed (Paper V [4]); the natural extensions are the higher-order kernels H_d, K_d through the same boundary problem (each order solves the same cohomological left-hand side with an explicit right-hand side), certified enclosures for the constructed values, and a self-contained verification of the classical strip geometry (K1) of the Kneser branch consumed there.

4. **Second structure and the x_* map.** Chart the indicated second singularity level at $y \approx 0.17$–0.22, and the pair map $x_*(b,d)$ (cheap: it is the accumulation point of the mode phases), including the suspiciously round measured centers.

5. **Edge and large-base asymptotics of the hub.** Push the edge fit window toward η to test the derived coefficient $\pi\sqrt{2\eta/e}$ below 0.5% (Lemma 8.2), and settle $m(b) - \ln\ln b \to 0$ with a rate.

6. **Foundations.** Derive the channel and peeling-growth hypotheses of [1] from the Kneser–Paulsen–Cowgill construction (the single remaining conditional element, cf. [2, Rem. 7.4]), including a stopping-rule formulation covering skew pairs; and prove Hypothesis 7.1 from base holomorphy [8].

Beyond these, the applied horizon that motivated the program remains: with the conversion maps certified, a single master engine in one base plus phase tables can, in principle, replace per-base Kneser constructions for multi-base workloads; whether the fixed-point formulation of $F_{b,d}$ converges fast enough in practice is an empirical question the hybrid format of Section 9 is designed to answer.

A Data tables

All values from the July 2026 campaign; run metadata in Appendix B.

A.1 Regenerated pair table (regular branch)

Means $\mu_{b,d}$ exact to the digits shown (25+ digits available in the repository); A_1, first-mode phase and A_2/A_1 from the dps = 80, $N = 256$ runs. This supersedes Table 1 of [1].

(b,d)	$\mu_{b,d}$	A_1	A_2/A_1
$(e,2)$	$-1.1284037766289240098792179$	1.4368×10^{-4}	0.00968
$(2,e)$	$+1.1284037766289240098792179$	1.4367×10^{-4}	0.00914
$(2,10)$	$+2.2649608387^*$	4.6080×10^{-4}	0.00865
$(10,2)$	-2.2649608387^*	4.6083×10^{-4}	0.01015
$(e,10)$	$+1.1365570557133264408993760$	3.1723×10^{-4}	0.00885
$(10,e)$	$-1.1365570557133264408993760$	3.1722×10^{-4}	0.00981
$(3,5)$	$+0.5788350021355712162374735$	—	—

(Starred values are hub-reconstructed, $\mu = m(d) - m(b) + \Delta$; the digits shown are limited by the two-digit published defect $\Delta(2,10) = 6.4 \times 10^{-9}$, while the full $\Delta^{(2)} + \Delta^{(3)}$ chain in the repository carries them to $\sim 10^{-15}$. Direct-run values (the e-pairs and $(3,5)$) are exact to the digits shown. The $(3,5)$ mode head resides in the repository; amplitudes for the closer pair are proportionally smaller by the amplitude law.)

A.2 Local decay rates, $(e,2)$, clean to $k = 290$

k	σ_{eff}	k	σ_{eff}	k	σ_{eff}
1	0.740	30	0.232	110	0.1689
2	0.567	40	0.2154	120	0.1657
3	0.487	50	0.2036	140	0.1601
5	0.404	60	0.1946	200	0.1484
8	0.344	70	0.1875	240	0.1429
20	0.260	80	0.1817	290	0.1375
		90	0.1768		
		100	0.1726		

Saddle-model reconstruction: $\sigma_{\text{eff}}(k) = \sigma_* + 1.090/\sqrt{k} - 1.512/k$ reproduces this table to 10^{-4} with the fitted parameters of Observation 6.2. Digit yield: $D(60) = 47$, $D(100) = 67$, $D(120) = 76$, $D(300) \approx 148$.

A.3 Breach zones at $y = 0.085$ (complexified ladder)

pair	zone x-interval	width	center x_* (mod 1)
$(e, 2)$	$[0.91, 0.98]$	≈ 0.07	≈ 0.945
$(3, 5)$	$[1.01, 1.08]$	≈ 0.07	≈ 1.045
$(2, 10)$	$[0.650, 0.710]$	≈ 0.06	≈ 0.680

Apex $(e, 2)$: $z_* \approx (1.02 \pm 0.02) + (0.055 \pm 0.005)i$; center slope $dx/dy \approx -2.8$; fitted singularity data (σ_*, c, γ): $(0.0789, 59.78, 9.47)$ for $(e, 2)$ and $(0.0791, 59.77, 9.59)$ for $(3, 5)$; phase strength ratio $c_{\text{phase}}/c_{\text{amp}} \approx 4.38$, $\chi \approx 0.68$ rad (pair-universal to 0.3%).

A.4 Hub raster and first-order constants

Hub values $m(b) = \mu_{e,b}$ as in Section 8; 24-point raster over $[1.46, 100]$ at 40 digits in the repository. Edge fit on $[1.46, 1.6]$: $m(b) = -3.3148\,(b - \eta)^{-1/2} - 0.398 \ln(b - \eta) + 3.163$ (rms 0.004); derived coefficient $\pi\sqrt{2\eta/e} = 3.23907$ (ratio 1.023). Kernel constants: $C_1(e) = 0.00038859729244516473$; $\overline{G}_e = 2.191463277066$; $C_1(d)\times 10^4 = 3.953, 3.886, 3.868, 3.793, 3.720$ for $d = 2, e, 3, 5, 10$; diagonal ratios $A_2/A_1 = 0.0093355030$, $A_3/A_1 = 2.66505\times 10^{-4}$, $A_4/A_1 = 1.26473 \times 10^{-5}$; diagonal phase $\varphi_1 = 1.33536363153$.

A.5 Branch-robustness budget

Mean shifts on the explicit C^1 branches versus the regular branch: $\Delta\mu = \mu_{\text{toy}} - \mu_{\text{reg}}$ equals $+6.37 \times 10^{-4}$ for $(e, 2)$, $+7.34 \times 10^{-4}$ for $(3, 5)$, $+3.758 \times 10^{-3}$ for $(e, 10)$, $+4.341 \times 10^{-3}$ for $(2, 10)$: empirically $|\Delta\mu| \approx (5\text{–}12) \times A_1(b, d)$—the choice of branch moves the mean by a few phase amplitudes, while the profile shapes remain similar. This quantifies the branch sensitivity of every constant in this paper.

B Reproducibility

Pipeline. Phases are evaluated by Algorithm 3.2 against the author's engine (fork of `fatou.gp` [9]; PARI/GP; engine, data, scripts and run metadata at `https://github.com/Lightrunnerwastaken/gp-tetration`; this paper is archived at `https://doi.org/10.5281/zenodo.21363593`, the companion papers at the DOIs given in the bibliography and the Authorship Statement), read-only, with per-point closure bounds logged; modes by direct DFT on N-point grids; the complexified scan by the bottom-up solver of Section 6 with path continuation and mid-strip cross-validation against the mode series.

Gates (all enforced per run). (i) truncation gate: one extra explicit tower level changes no phase value by more than 10^{-120}; (ii) mean antisymmetry $\leq 10^{-39}$ pairwise; (iii) groupoid residual $\leq 2.4\times 10^{-34}$ (storage-limited); (iv) toy-branch control: measured N^{-3} aliasing against the independent value of [2, Rem. 7.5], and inverse identity to 5.7×10^{-101}; (v) instrument guard $k \leq 2.3\, n_{\text{lim}}$.

Runs (selection).

tag	pairs	dps	N	K	notes
toy	$(e,2),(2,e),(2,10),(e,10),(3,5)$	mp100	512	64	calibration, $p \approx 3$
reg	all 12 regular pairs	80	256	64	cache-hot, < 5 s each
deep	$(e,2)$	200	512	176	floor study, $n_{\text{lim}} = 30/60$
edge	$(1.5,e),(e,1.5)$	60	256	48	init 1.2 s
decisive	$(e,2)$, $n_{\text{lim}} = 120$	190	1024	300	inits 239 s + 196 s
decisive	$(3,5)$, $n_{\text{lim}} = 120$	190	1024	300	tail universality
diag	(d',e), $\ln d' = 1+\varepsilon$	200	512	—	$\varepsilon = 10^{-12}, 10^{-15}$

Authorship and AI Contribution Statement

This manuscript is the product of a directed human–AI collaboration, and the division of labor should be stated plainly. The author conceived the mixed-base phase program and its central object $\Phi_{b,d}$, built and maintains the high-precision tetration engine (a fork of Levenstein's `fatou.gp`, published at https://github.com/Lightrunnerwastaken/gp-tetration) against which every number in this paper was computed, and directed the research: posing the questions, setting priorities between rounds, auditing intermediate results, and deciding which claims met the acceptance gates. The majority of the mathematical development and of the numerical work was produced by artificial-intelligence systems under that direction: an interactive assistant (Claude, Anthropic) contributed the theoretical framework of Sections 3, 4, 6–8 (including the certificate propositions, the saddle-point and crossover analysis, the response-kernel theory, the third-order hub defect and the parabolic edge law), the round-by-round experimental designs, and the review of the companion manuscripts; an autonomous research agent designed, implemented and executed the measurement pipeline, discovered and quantified the instrument systematics, performed the complexified scans, and produced all data reported here. The companion manuscripts [1, 2] were developed in the same directed mode with a further AI system (ChatGPT, OpenAI) as the principal drafting contributor. In line with prevailing journal policies, AI systems are not listed as authors; the author assumes full and sole responsibility for the correctness of all statements, and readers should weigh the provenance described here in their assessment. The five papers of the series are permanently archived on Zenodo under the concept DOIs https://doi.org/10.5281/zenodo.21346803 (I), https://doi.org/10.5281/zenodo.21361967 (II), https://doi.org/10.5281/zenodo.21362206 (III), https://doi.org/10.5281/zenodo.21363593 (IV), and https://doi.org/10.5281/zenodo.21363809 (V). The author's role, in short, was closer to that of a research director than of a sole working mathematician; the author considers it a matter of scientific honesty that this be recorded rather than obscured.

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